

B.Tech Degree V Semester Examination December 2004

ME 505 HEAT AND MASS TRANSFER

(2002 Admissions)

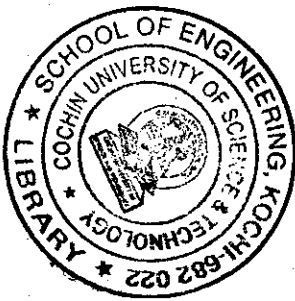
Time: 3 Hours

Maximum Marks: 100

(Use of HMT Data book permitted)

- I. (a) Explain the concept of overall heat transfer coefficient and derive an expression for the same for a plane wall. (8)
- (b) A furnace wall is made of 75 mm thick fire clay brick and 6.4 mm thick mild steel plate. The inside surface of brick is at 650°C and the out side air is at 30°C. Determine –
 (i) The heat loss per m² area of the furnace wall and out side surface temperature of steel plate. (ii) If 18 mild steel bolts each of 19 mm diameter are used to fix the steel plate on the brick wall per m² area of the wall, what % increase in the heat flux will result? Thermal conductivity of Brick = 1.1 W/mk. Mild steel = 39 W/mk. Air side film heat transfer coefficient = 68 W/m²k. (12)
- OR**
- II. (a) Explain the role of Biot number in extended surface heat transfer. (5)
- (b) An electrical current of 34000 amps flows along a flat steel plate 1.25 cm thick and 10 cm wide. The temperature of one surface of the plate is at 80°C and that of the other is at 95°C. Find the temperature distribution across the plate and the value and position of maximum temperature. Also calculate total amount of heat generated per metre length of plate and flow of heat from each side of the plate –
 Resistivity of steel $\rho = 12 \times 10^{-6} \Omega \text{ cm}$.
 Thermal conductivity = 54 W/m K. (15)
- III. (a) Explain the significance of non dimensional numbers in natural and forced convection heat transfer. (8)
- (b) Saturated steam under atmospheric pressure condenses on a vertical surface 100 cm high. The temperature of the surface is 80°C. At 50 cm from the top of the plate, calculate –
 (i) Film thickness
 (ii) The local heat transfer coefficient
 (iii) Average heat transfer coefficient for 50 cm height of the plate
 (iv) Rate of condensation per hour (12)
- OR**
- IV. (a) Derive an expression for film thickness in film wise condensation on a vertical flat plate. (8)
- (b) A long pipe 15 cm outer diameter having surface temperature 80°C passes through a water tank. The water is at 40°C. Calculate the heat transfer per unit length of the tube. Properties of water can be taken as –
 $\rho = 983.2 \text{ Kg/m}^3$, $\mu = 4.59 \times 10^{-3} \text{ NS/m}^2$, $C_p = 4.186 \text{ KJ/Kg K}$.
 $K = 0.634 \text{ w/mk}$. Use the following relation $Nu = 0.525 Ra^{0.25}$. (12)
- V. (a) Explain the following :
 (i) Kirchoff's Law (ii) Wien's displacement Law
 (iii) Solid angle and intensity of radiation (9)

(Turn Over)



- (b) Two very large parallel plates with emissivities 0.4 and 0.75 are engaged in radiative heat transfer. Calculate the % reduction in heat transfer when three highly polished aluminum surfaces with emissivity 0.04 are placed in between them. (11)

OR

- VI. (a) Explain the network analysis of radiation heat transfer problem. (8)
 (b) A glass plate 40 cm x 40 cm in size is used on an electric furnace to view the radiation. The transmissivity of the glass is 0.5 in the view b and of $0.2\mu\text{m}$ to $4\mu\text{m}$. The emissivity is 0.3 up to $4\mu\text{m}$ and 0.9 above that. The transmissivity of the glass is zero except in the range from 0.2 to $4\mu\text{m}$. Calculate the energy absorbed and transmitted by the glass assuming the furnace is a black body at 2000K. (12)
- VII. (a) Explain the NTU method of heat exchanger design. (8)
 (b) Water at the rate of 4Kg/sec is heated from 35°C to 55°C in a shell and tube heat exchanger. On the shell side one pass is used with water on the heating fluid with flow rate of 2Kg/sec entering the heat exchanger at 95°C . Take $U = 1420\text{ w/m}^2\text{K}$ and the average water velocity is the 20 mm diameter tube is 0.4 m/sec. Because of the space limitation the tube length must not be longer than 2.5 m. Calculate the number of tube passes, the number of tubes per pass and the minimum length of the tubes considering the flow as counter flow. (12)
- OR
- VIII. (a) What is meant by fouling, how it affects the performance of heat exchangers ? Explain. (8)
 (b) Water to water heat exchanger of a counter flow type has heating surface area of 2m^2 . The other heat exchanger parameters are as follows:
 $T_{hi} = 85^\circ\text{C}$, $m_h = 2000\text{ kg/hr}$, $T_{ci} = 25^\circ\text{C}$, $m_c = 1500\text{ kg/hr}$, $U = 1400\text{ w/m}^2\text{K}$. Determine the amount of heat transfer and the outlet temperatures of the cold and hot fluids. (12)

- IX. (a) Bring out the analogy between Mass, Momentum and Heat transfer. (8)
 (b) Water has been spilled on the floor, the water layer is 1 mm and at 25°C . The air is also at 25°C and 1 atm. The absolute humidity of air is 2 gm/kg of dry air. Determine the time required to evaporate the complete water if the evaporation takes place by molecular diffusion through an air film of 5 mm thick. (12)

OR

- X. (a) Explain equimolar diffusion. (8)
 (b) Dry air at atmospheric pressure blows across a thermometer which is enclosed in a dampened cover. The thermometer reads a temperature of 18.3°C , determine the temperature of the dry air. (12)
